# Digital Task Intensity of Indian Occupations — results and status

Nair, Anand & Shome (Nirma University) · SAU Digital Markets · status at 21 October 2026, 09:58 IST
For co-authors. All numbers come from the result files in the project repository (`dti-india`).

> **In one paragraph.** The LLM instrument has scored all 3,450 NCO-2015 occupations and is stable and grounded in the official text. Linked to 466,095 PLFS wage earners (2017-18 to 2023-24), a one-standard-deviation higher India-specific DTI goes with **4.1 log points (≈ 4.2%) higher hourly wages** within the same occupational division (SE 1.3), somewhat below the 5.7 obtained with the US O*NET measure. The premium is concentrated at the top of the wage distribution and is absent for workers with at most primary education, informal regular jobs and casual work — the pattern the abstract calls a *digital-returns gap*. It is a conditional association, not a causal return. Still to come: the second model (today, ≈ 16:00), the India–US comparison RQ2 (tonight) and human validation (forms due tonight).

## 1. What is running and what is left

| Step | Status | Detail |
|---|---|---|
| Qwen2.5-14B scores all 3,450 NCO occupations × 5 runs | Done (21 Oct, 05:50) | 17,250 / 17,250, 0 failed |
| Wage analysis with the NCO-LLM DTI (RQ3–RQ4) | Done (21 Oct, 08:34) | 10 tables |
| Llama-3.1-8B scores all occupations (second model, cross-check) | Running | 41% · finishes ≈ 16:00, 21 Oct |
| Qwen2.5-14B scores 923 O*NET occupations (RQ2 same-instrument benchmark) | Queued | finishes ≈ 00:00, 22 Oct |
| Human validation: 3 coders × 200 occupations | With coders | forms due 21 Oct, 22:00 |

- Jobs run on PARAM Shavak (Quadro GP100) inside `screen`; they do not depend on anyone's laptop.
- After the queue finishes: aggregate both models, cross-model agreement, India vs O*NET gap (RQ2), wage robustness with the second model, then the human–LLM comparison once forms pass the quality checks. About half a day of work; updated paper draft on 22 Oct.
- Decisions still open for the professors: wage-model choices marked OPEN in `configs/analysis.yaml`; confirmation of the strict rubric (Section 2).

## 2. RQ1 — Is the LLM a reliable measurement instrument?

**Set-up.** Main instrument: Qwen2.5-14B-Instruct, strict rubric (score only digital work the NCO-2015 text names; silent → 0), five independent runs per occupation at temperature 0.7, median across runs. Five dimensions scored 0–4; DTI = equal-weight sum of the z-scored dimensions. The strict rubric was chosen in the pilot (50 occupations, two models, two rubric readings): the broader "described-task" reading barely reduced the share of all-zero occupations (48% vs 50%) and lowered agreement between models (κ 0.69 vs 0.83).

### Stability across repeated runs (all 3,450 occupations)

| Dimension | ICC(2,1) | Krippendorff α | All 5 runs identical | Occupations scored > 0 |
|---|---|---|---|---|
| Basic digital operation | 0.73 | 0.72 | 62% | 33% |
| Digital information processing | 0.82 | 0.81 | 72% | 28% |
| Digital communication | 0.70 | 0.65 | 90% | 5% |
| Specialised digital tools | 0.81 | 0.79 | 60% | 36% |
| Digital problem solving / content | 0.84 | 0.79 | 84% | 13% |

**Meaning.** ICC/α of 0.70–0.84 is "good" agreement by the usual benchmarks (0.75+ good, 0.6–0.75 moderate): asked five times, the model gives largely the same scores. Communication (D3) is the least stable because it is rarely non-zero, so a single deviating run weighs heavily. 1,190 occupations (34%) vary by more than 0.5 points (SD) on at least one dimension; they are listed for review, and the median across five runs is used precisely to absorb this noise.

### Grounding and consistency

- **87%** of the 20,355 non-zero dimension scores come with a quote that is found in the official description (pilot: 89%). The rest are mostly scores without a quote, where the model drew on general knowledge despite the rule.
- Equal-weight dimension mean vs the model's own holistic 0–10 judgement (Q9): Spearman **0.84** — supports equal weighting.
- Correlation with a simple keyword count on the same text: Spearman **0.30**. Occupations whose description names a digital term average 1.56 (0–4) vs 0.39 for the rest; only 7% of the former are all-zero. The LLM therefore picks up much more than keywords, but stays tied to the text.
- **51.6%** of occupations score 0 on every dimension. This is a finding about the NCO-2015 text, not an error: only about 5% of descriptions use any digital term. It is the main reason to compare with O*NET (RQ2).

### Mean score (0–4) by division

| Division | Occ. | Mean | All-zero |
|---|---|---|---|
| 1. Managers | 180 | 0.30 | 68% |
| 2. Professionals | 647 | 1.05 | 31% |
| 3. Technicians & assoc. professionals | 546 | 0.65 | 37% |
| 4. Clerical support | 118 | 0.59 | 35% |
| 5. Service & sales | 154 | 0.28 | 66% |
| 6. Skilled agriculture | 119 | 0.05 | 89% |
| 7. Craft & related trades | 859 | 0.20 | 69% |
| 8. Plant & machine operators | 715 | 0.26 | 44% |
| 9. Elementary occupations | 112 | 0.06 | 88% |

### Most digital occupations

| Occupation | Mean |
|---|---|
| Programmer Analyst | 3.2 |
| Software Engineer | 3.2 |
| Programmer, Engineering and Scientific/System Programmer | 3.0 |
| Astronomer | 2.8 |
| Caption Writer | 2.8 |
| Radio Engineer, Telecommunication | 2.8 |
| Web Designer | 2.8 |
| Associate-Analytics | 2.6 |
| Computer Programmer/Software Engineer | 2.6 |
| Computer Programmers, Other | 2.6 |

**Meaning.** The ranking is economically sensible (professionals > technicians > clerks > … > agriculture and elementary work) and the most digital occupations are the expected ones — face validity for the instrument. PLFS match rate: 98.7–99.8% of employed workers in each round receive a DTI.

> **Pending for RQ1.** (i) Agreement between the two model families on all occupations (Llama finishes today; pilot: Spearman 0.77, weighted κ 0.83). (ii) Human validation: 3 coders × 200 occupations (stratified by division and by whether the text names digital work; the pilot occupations excluded). The first round of coder forms was not usable (identical scores on all five dimensions for every occupation).

## 3. RQ3 — Conditional digital occupational wage premium

ln(real hourly wage) of regular and casual wage earners aged 15–64, seven PLFS rounds, PLFS weights, standard errors clustered by occupation (233 occupation groups). Coefficients in **log points per one standard deviation of DTI** (≈ % difference in wages). SE in brackets; *** p<0.01, ** p<0.05, * p<0.1.

| Specification | NCO-LLM DTI (India) | $R^2$ | O*NET-transferred DTI |
|---|---|---|---|
| **M1** No controls | 27.5*** (3.2) | 0.24 | 30.4*** (2.5) |
| **M2** + individual characteristics | 8.5*** (2.6) | 0.46 | 11.2*** (1.9) |
| **M3** + state, industry, year-quarter FE | 6.3*** (1.3) | 0.56 | 10.7*** (1.6) |
| **M4** + occupational division FE (preferred) | 4.1*** (1.3) | 0.57 | 5.7*** (1.5) |

**Meaning.** Without controls, more digital occupations pay 27.5 log points more per SD. Most of that reflects who works in them (education, age, location, industry). Comparing occupations within the same broad division, the premium is **4.1 log points (≈ 4.2%)**: modest but clearly different from zero (p = 0.002). With the US O*NET measure it is 5.7: transferring US task content gives a somewhat larger premium.

### How robust is it?

| Check (all M4) | NCO-LLM DTI | O*NET DTI |
|---|---|---|
| M4 unweighted | 3.5*** (1.3) | 6.4*** (1.6) |
| M4 unweighted, wild cluster bootstrap | 3.5** (1.3) | 6.4*** (1.6) |
| M4 monthly earnings | 5.2*** (1.5) | 8.4*** (1.4) |
| M4 regular wage workers only | 3.9*** (1.3) | 5.0*** (1.5) |
| M4 casual workers only | -1.6 (1.8) | 0.9 (1.8) |
| M4 NCO-2015 rounds (Jul 2021+) | 4.6** (2.1) | 4.4*** (1.3) |
| M4 NCO-2004 rounds (pre Jul 2021) | 3.1* (1.6) | 6.4*** (2.0) |
| M4 NCO-2004 groups with ≥75% direct links | 5.4*** (1.8) | 5.5*** (1.5) |
| M4 self-employed (earnings incl. capital returns) | 0.5 (1.1) | -3.9* (2.2) |
| Heckman selection correction (all) | 4.2*** (1.3) | 5.7*** (1.4) |
| DTI quintile 2 vs lowest fifth | 3.9 (2.5) | 0.3 (2.0) |
| DTI quintile 3 vs lowest fifth | 1.5 (2.5) | 2.5 (2.1) |
| DTI quintile 4 vs lowest fifth | 8.3*** (2.8) | 3.6 (2.5) |
| DTI quintile 5 vs lowest fifth | 9.3*** (2.9) | 13.5*** (4.1) |

**Meaning.** The premium survives unweighted estimation, the wild cluster bootstrap, monthly earnings, direct links only and correction for selection into wage work; it is weaker (significant only at 10%) in the pre-2021 NCO-2004 rounds. It is **not linear**: only the two most digital fifths of occupations earn significantly more. It is absent for casual workers and the self-employed.

> **Not causal.** Oster (2019) $\delta = 0.34$ (O*NET: 0.40): selection on unobservables only about a third as strong as selection on observables would explain the premium away. Workers sort into digital occupations by ability and background, so the paper reports β as a **conditional occupational association**, as stated in the abstract.

## 4. RQ4 — Who gains? Heterogeneity

### Across the wage distribution (unconditional quantile / RIF regressions)

| Wage percentile | NCO-LLM DTI | O*NET DTI |
|---|---|---|
| 10th percentile | -1.7 (1.3) | 1.6 (1.5) |
| 25th percentile | -0.2 (1.3) | 1.8 (1.5) |
| 50th percentile | 2.1 (1.4) | 3.6** (1.8) |
| 75th percentile | 6.2*** (1.7) | 7.8*** (2.3) |
| 90th percentile | 21.3*** (6.7) | 19.8*** (6.1) |

**Meaning.** No premium in the lower half of the distribution; it rises to about 21.3 log points at the 90th percentile. Occupational digital intensity pays mainly at the top.

### By group (M4 estimated separately for each group)

| Group | NCO-LLM DTI | O*NET DTI |
|---|---|---|
| Education: up to primary | -2.1 (1.5) | -0.4 (2.3) |
| Education: middle / secondary | 3.5*** (1.2) | 4.8*** (1.5) |
| Education: tertiary | 4.7*** (1.5) | 7.3*** (1.8) |
| Regular, no social security (informal) | 1.0 (0.8) | -0.3 (1.5) |
| Regular, with social security (formal) | 3.8*** (1.2) | 5.6*** (1.6) |
| Regular wage workers | 3.9*** (1.3) | 5.0*** (1.5) |
| Casual workers | -1.6 (1.8) | 0.9 (1.8) |
| Men | 3.6*** (1.2) | 3.5** (1.4) |
| Women | 5.6** (2.2) | 8.6*** (2.0) |
| Urban | 3.1** (1.3) | 4.0*** (1.4) |
| Rural | 4.6*** (1.7) | 7.8*** (1.9) |
| Age 15–24 | 2.5** (1.1) | 2.6** (1.2) |
| Age 45–54 | 4.3** (1.9) | 8.3*** (2.2) |

### Within the same 3-digit occupation (interactions with full occupation fixed effects)

| Difference in the premium | NCO-LLM DTI | O*NET DTI |
|---|---|---|
| Women vs men | 3.7*** (0.7) | 3.2*** (1.0) |
| Rural vs urban | -3.0*** (0.6) | -3.4*** (0.7) |
| Middle/secondary vs up to primary | 8.6*** (2.4) | 8.9*** (1.7) |
| Tertiary vs up to primary | 11.5*** (3.3) | 18.0*** (2.1) |
| Casual vs regular | -5.9** (2.7) | -0.2 (2.2) |
| Formal vs informal (regular workers) | 4.0*** (1.4) | 11.0*** (2.0) |

**Meaning.**

- **Education is the clearest divide.** No premium for workers with at most primary schooling (-2.1, n.s.); 4.7 for graduates. Within the same occupation, graduates gain about 12 log points more per SD than the least educated.
- **Formality.** Among regular workers, the premium exists only with social-security coverage (3.8 vs 1.0, n.s.). Casual workers get none.
- **Gender.** Women earn a larger premium than men (5.6 vs 3.6; machine-learning group effects 6.3 vs 2.9), but occupational DTI explains almost none of the 31.7-point gender wage gap (Oaxaca–Blinder: 0.5 points; O*NET: -0.1). Most of the gap (28.0 points) is unexplained.
- **Rural–urban is mixed.** Estimated separately, the premium is higher for rural workers; within the same occupation it is lower. This needs care in the paper and is reported as inconclusive.
- **Machine-learning check** (double/debiased ML, 200,000-worker subsample): average effect 3.7 log points (SE 0.3); O*NET 3.6. Group effects confirm the education and gender gradients.

> **Policy reading (abstract, Section 7).** Digital intensity of an occupation translates into higher pay mainly for educated, formally employed, higher-paid workers. For the less educated, informal and casual workforce the premium is zero. This is a **digital-returns gap**, not only a digital-skills gap: basic digital training alone is unlikely to close it without formalisation, advanced skills and access to digitally enabled jobs.

## 5. RQ2 — India-specific vs O*NET-transferred intensity

Partly visible already: with the same wage model, the US O*NET measure gives a larger average premium (5.7 vs 4.1) and a larger gradient by education and formality. Whether this is because US occupations are described as more digital than their Indian counterparts is exactly what the RQ2 comparison tests. **Pending tonight:** the same LLM prompt applied to O*NET texts (so the gap reflects texts, not method), correlations at the occupation-family level, and the DigitalGap by division and skill level.

## 6. Data and reproducibility

- NCO-2015 Vol II descriptions (3,450 occupations); PLFS 2017-18 to 2023-24 (seven annual rounds, CWS wages, MoSPI weights), pre-2021 rounds linked through an NCO-2015 → NCO-2004 concordance (2,657 direct links, others via siblings); O*NET 31.0 with BLS SOC–ISCO crosswalk and OEWS weights.
- LLMs run locally (llama.cpp, GGUF): no occupation text leaves the university server. Prompt and rubric are versioned (hash 46a323537a).
- PLFS microdata are not redistributed; only estimates appear in the repository. Code: private GitHub repository `dti-india` (guide for writing: `docs/PAPER_GUIDE.md`).

Preliminary results for internal discussion; numbers may change slightly once the second model and the human validation are added.